

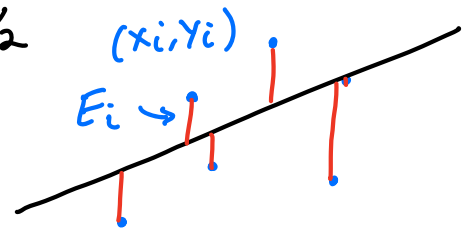
9-25

Least Squares Regression line

Def 9.1 The least squares regression line to the data set $\{(x_i, y_i) : i=1, \dots, n\}$ is the equation $y = \hat{a}x + \hat{b}$ such that the error

$$E = \left(\sum_{i=1}^n \underbrace{(y_i - (ax_i + b))^2}_{\sum E_i^2} \right)^{1/2}$$

is minimized.



Theorem 9.2 Given the data set $\{(x_i, y_i) : i=1, \dots, n\}$ the least squares regression line is given by $y = \hat{a}x + \hat{b}$ where

$$\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix}^{-1} \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}$$

Proof : Consider the linear system

$$ax_1 + b = y_1$$

$\vdots$

$$ax_n + b = y_n$$

which is rewritten as

$$\begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Now the least squares solution is

$$\begin{aligned} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} &= (A^T A)^{-1} A^T \vec{b} \\ &= \left(\begin{bmatrix} x_1 & \dots & x_n \\ 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} x_1 & \dots & x_n \\ 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \\ &= \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix}^{-1} \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix} \end{aligned}$$

Note that $\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix}$ is invertible provided

that $(\sum x_i)^2 \neq \sum x_i^2$

Coefficient of Determination (R^2)

let $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ denote the mean. Then

$$R^2 = 1 - \frac{E^2}{\sum_{i=1}^n (\bar{y} - y_i)^2}$$

measures goodness of fit

Exercise:

- 1) Find the line of best fit to the data set $\{(-1, 1), (0, -2), (1, 4), (2, 10)\}$ and compute R^2 .

$$\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix}^{-1} \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}$$

$$\text{Setting } \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 23 \\ 13 \end{bmatrix}$$

$$= \frac{1}{20} \begin{bmatrix} 4 & -2 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} 23 \\ 13 \end{bmatrix}$$

$$= \frac{1}{20} \begin{bmatrix} 66 \\ 32 \end{bmatrix}$$

We have that $y = \frac{33x}{10} + \frac{8}{5}$ is the line of best fit.

R^2 :

i	x_i	$(\bar{y} - y_i)^2$	$(3.3x_i + 1.6 - y_i)^2$
1	-1	$(\frac{9}{4})^2$	$(2.7)^2$
2	0	$(\frac{21}{4})^2$	$(3.6)^2$
3	1	$(\frac{3}{4})^2$	$(0.9)^2$
4	2	$(\frac{27}{4})^2$	$(1.8)^2$

$$\begin{aligned} \bar{y} &= \frac{\sum y_i}{4} \\ &= \frac{13}{4} \end{aligned}$$

$$R^2 = 1 - \frac{(2.7)^2 + (3.6)^2 + (0.9)^2 + (1.8)^2}{\left(\frac{9}{4}\right)^2 + \left(\frac{21}{4}\right)^2 + \left(\frac{3}{4}\right)^2 + \left(\frac{27}{4}\right)^2} \approx 0.69$$

Graham - Schmidt

Recall: The G-S algorithm inputs a set of linearly independent vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ and outputs a set of orthonormal vectors $\{\vec{e}_1, \dots, \vec{e}_n\}$ such that $\text{span}(\vec{v}_1, \dots, \vec{v}_k) = \text{span}(\vec{e}_1, \dots, \vec{e}_k)$ for all $k = 1, \dots, n$.

Step 1: Set $\vec{e}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$

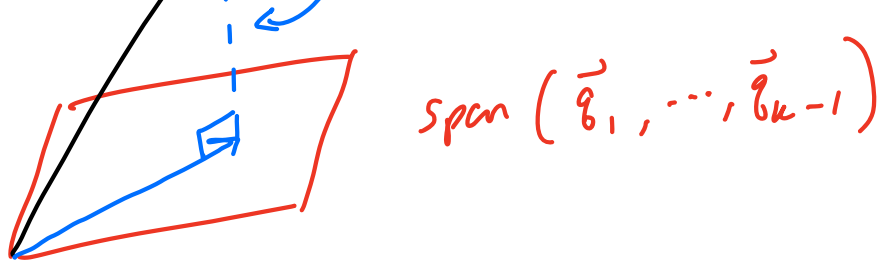
Step 2: Set $\vec{x}_2 = \vec{v}_2 - \langle \vec{v}_2, \vec{e}_1 \rangle \vec{e}_1$ and let $\vec{e}_2 = \frac{\vec{x}_2}{\|\vec{x}_2\|}$

⋮

Step k : Set $\vec{x}_k = \vec{v}_k - \sum_{i=1}^{k-1} \langle \vec{v}_k, \vec{e}_i \rangle \vec{e}_i$ and let $\vec{e}_k = \frac{\vec{x}_k}{\|\vec{x}_k\|}$

Note that $\sum_{i=1}^{k-1} \langle \vec{v}_k, \vec{e}_i \rangle \vec{e}_i$ is the orthogonal projection of $\vec{v}_k$ onto $\text{span}(\vec{e}_1, \dots, \vec{e}_{k-1})$





Exercise:

2) Given $\vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ find orthonormal vectors $\{\vec{q}_1, \vec{q}_2, \vec{q}_3\}$ such that $\text{span}(\vec{q}_1, \vec{q}_2) = \text{span}(\vec{v}_1, \vec{v}_2)$ and $\text{span}(\vec{q}_1) = \text{span}(\vec{v}_1)$

Using G-S we set

$$\vec{q}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

and $\vec{q}_2 = \frac{\vec{x}_2}{\|\vec{x}_2\|}$ where

$$\vec{x}_2 = \vec{v}_2 - \langle \vec{v}_2, \vec{q}_1 \rangle \vec{q}_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} - \left(\begin{bmatrix} 1 & 1 & -1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right) \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\|c\vec{v}\| = |c| \|\vec{v}\|$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 3 \\ 5 \\ -6 \end{bmatrix}$$

$$\text{So } \vec{q}_2 = \frac{\frac{1}{5} \begin{bmatrix} 3 \\ 5 \\ -6 \end{bmatrix}}{\frac{1}{5} \sqrt{70}} = \frac{1}{\sqrt{70}} \begin{bmatrix} 3 \\ 5 \\ -6 \end{bmatrix}$$

Setting $\vec{q}_3 = \vec{q}_1 \times \vec{q}_2$ we arrive at the desired set as

$$\begin{aligned} \text{Span}(\vec{q}_1, \vec{q}_2) &= \text{Span}(\vec{v}_1, \vec{v}_2) \\ &\begin{cases} \vec{v}_1 = \sqrt{5} \vec{q}_1, & \Rightarrow \text{Span}(\vec{v}_1) = \text{Span}(\vec{q}_1) \\ \vec{v}_2 = \|\vec{x}_2\| \vec{q}_2 + \langle \vec{v}_2, \vec{q}_1 \rangle \vec{q}_1 \\ &= \frac{\sqrt{70}}{5} \vec{q}_2 + \frac{1}{\sqrt{5}} \vec{q}_1 \end{cases} \end{aligned}$$

Note that if we only need to obtain orthogonal (not nec. unit) vectors, we can modify G-S as follows

$$\text{step 1: set } \vec{x}_1 = \vec{v}_1$$

$$\text{step 2: set } \vec{x}_2 = \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{x}_1 \rangle}{\langle \vec{x}_1, \vec{x}_1 \rangle} \vec{x}_1$$

⋮

$$\text{step } k: \text{ set } \vec{x}_k = \vec{v}_k - \sum_{i=1}^{k-1} \frac{\langle \vec{v}_k, \vec{x}_i \rangle}{\langle \vec{x}_i, \vec{x}_i \rangle} \vec{x}_i$$

orthogonal projection of $\vec{v}_k$

onto $\text{span}(\vec{x}_1, \dots, \vec{x}_{k-1})$

Exercises

- 3) Consider $P_3(\mathbb{R})$ endowed with the inner product $\langle f, g \rangle = \int_{-1}^1 f \cdot g \, dx$. Use G-S to construct an orthogonal basis.

Letting $e_i = x^i$, $i = 0, 1, 2, 3$ we have that

$$\langle e_i, e_j \rangle = \int_{-1}^1 x^{i+j} = \begin{cases} 0 & \text{if } i+j = 2n+1 \\ \frac{2}{i+j+1} & \text{if } i+j = 2n \end{cases}$$

Now applying G-S we obtain

$$f_0 = e_0 = 1$$

$$f_1 = e_1 - \frac{\langle e_1, f_0 \rangle}{\langle f_0, f_0 \rangle} f_0 = x - \frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} 1 = x$$

$$f_2 = e_2 - \frac{\langle e_2, f_0 \rangle}{\langle f_0, f_0 \rangle} f_0 - \frac{\langle e_2, f_1 \rangle}{\langle f_1, f_1 \rangle} f_1$$

$$\begin{aligned}
 &= x^2 - \frac{\langle x^2, 1 \rangle}{\langle 1, 1 \rangle} 1 - \frac{\langle x^2, x \rangle}{\langle x, x \rangle} x \\
 &= x^2 - \frac{2/3}{2} = x^2 - \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 f_3 &= e_3 - \frac{\langle e_3, e_0 \rangle}{\langle e_0, e_0 \rangle} e_0 - \frac{\langle e_3, e_1 \rangle}{\langle e_1, e_1 \rangle} e_1 \\
 &\quad - \frac{\langle e_3, x^2 - \frac{1}{3} \rangle}{\langle x^2 - \frac{1}{3}, x^2 - \frac{1}{3} \rangle} (x^2 - \frac{1}{3}) \\
 &= x^3 - \frac{2/5}{2/3} x = x^3 - \frac{3x}{5}
 \end{aligned}$$

$\therefore$ The orthogonal basis is $\left\{ 1, x, x^2 - \frac{1}{3}, x^3 - \frac{3x}{5} \right\}$